

Ecosystem modeling: Individual-based model

III. Adding biological movement

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Recap

One of the simplest (but powerful) individual-based model is particle tracking algorithm:

(Simple) particle tracking algorithm with IBM

$$\frac{dX_i}{dt} = U_i \quad (1a)$$

$$\frac{dY_i}{dt} = V_i \quad (1b)$$

$$\frac{dQ_i}{dt} = \sum SS \quad (1c)$$

Source & Sink

where

X_i and Y_i : x - and y -position of i -th individual (particle)

U_i and V_i : each velocity component at the position (X_i, Y_i) .

Q_i : Biological/physiological quantity of interest

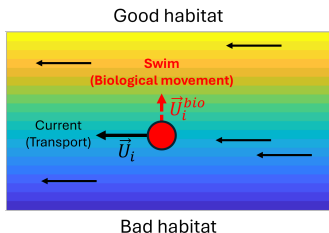
Biological movement (swimming ability)

Particle tracking with directional swimming

$$\frac{dX_i}{dt} = U_i + U_i^{bio}(Q) \quad (2)$$

$$\frac{dY_i}{dt} = V_i + V_i^{bio}(Q) \quad (3)$$

where U_i^{bio} and V_i^{bio} indicate vector component in each direction caused by swimming ability of an individual, that depends on quality of habitat Q .



Practical example: Choi et al. (2026)

Step 1: Model and setup quality of habitat Q .

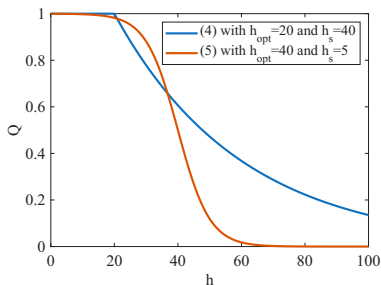
A knowledge: Lobsters habitat is usually shallower than 20 m depth. So, we may model quality of habitat using below formulation (Choi et al., 2026):

$$Q = \min \left[e^{-(h-h_{opt})/h_s}, 1 \right] \quad (4)$$

Of course, there are tons of alternative formulation such as

$$Q = \frac{1}{1 + e^{-(h-h_{opt})/h_s}} \quad (5)$$

where h_{opt} and h_s are constant parameters.



Practical example: Choi et al. (2026)

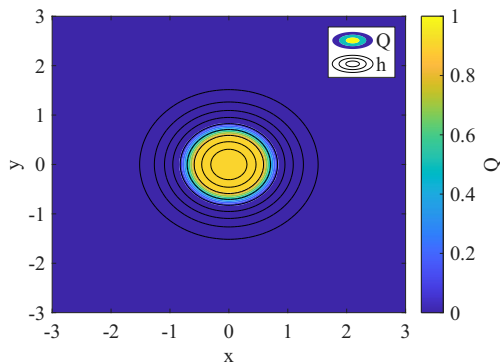
Step 1: Model and setup quality of habitat Q .

Let us consider below toy example:

$$h = h_0 - h' e^{-(x^2+y^2)/R^2} \quad (6)$$

$$Q = \frac{1}{1 + e^{-(h-h_{opt})/h_s}} \quad (7)$$

where $h_0 = 1$, $h' = 0.99$, $R = 1$, $h_{opt} = 0.4$, and $h_s = 0.05$.



Practical example: Choi et al. (2026)

Step 2: Setting $\vec{u}^{bio}$ letting them move toward a region where higher Q .

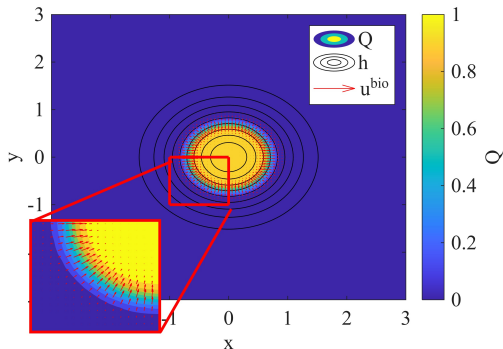
How? Use gradient!

$$\vec{u}^{bio} = (u^{bio}, v^{bio}) \sim \nabla Q = \left(\frac{\partial Q}{\partial x}, \frac{\partial Q}{\partial y} \right) \quad (8)$$

Vector field that “uphills” Q

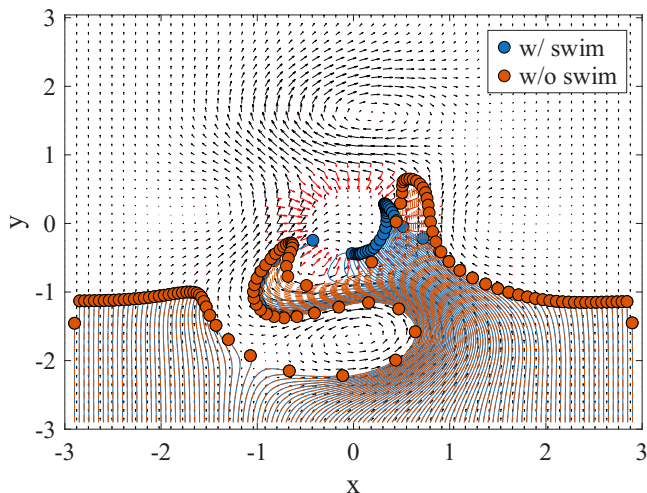
For simplicity, we may consider simple formulation:

$$\vec{u}^{bio} = \alpha \nabla Q \quad (9)$$



Practical example: Choi et al. (2026)

Step 3: Do particle tracking on $\vec{u} + \vec{u}^{bio}$!



Nondirectional movement (random walk)

Particle tracking with directional swimming

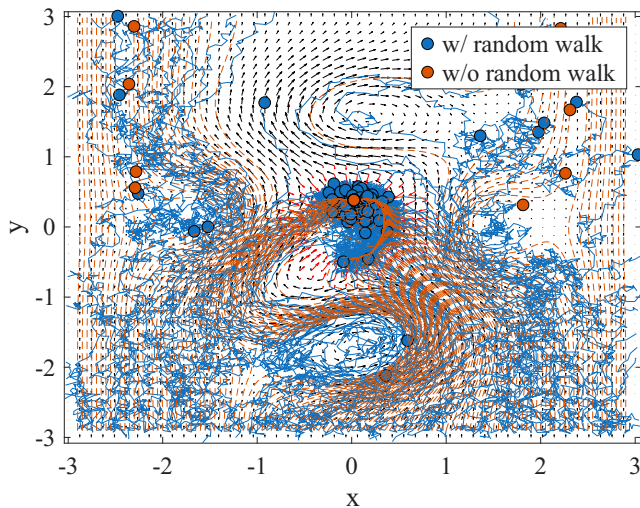
$$X_i^{n+1} = X_i^n + \Delta t(U_i + U_i^{bio})^* + N(0, \sqrt{2A\Delta t}) \quad (10)$$

$$Y_i^{n+1} = Y_i^n + \Delta t(V_i + V_i^{bio})^* + N(0, \sqrt{2A\Delta t}) \quad (11)$$

$$\approx \frac{dY_i}{dt} = V_i + V_i^{bio}$$

where $N(\mu, \sigma)$ indicates normal random number whose mean and standard deviation are μ and σ , respectively. The parameter A , determining magnitude of the random walk, can be referred to as **diffusion** coefficient.

Nondirectional movement (random walk)



Nondirectional movement (random walk)

Note that random walk of individual is diffusion of population¹:

$$\begin{aligned}X_i^{n+1} &= X_i^n + \Delta t(U_i + U_i^{bio})^* + N(0, \sqrt{2A\Delta t}) \\Y_i^{n+1} &= Y_i^n + \Delta t(V_i + V_i^{bio})^* + N(0, \sqrt{2A\Delta t})\end{aligned}\quad (12)$$

(Individual-based model)



$$\frac{\partial C}{\partial t} + \frac{\partial((u + u^{bio})C)}{\partial x} + \frac{\partial((v + v^{bio})C)}{\partial y} = A \left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} \right)$$

(Population dynamics model)

(13)

¹https://jang-geun.github.io/vis_geo_adv_diff_1.gif

References I



Choi, Jang-Geun et al. (2026). “Modeling the Influence of Directional Swimming Ability in American Lobster (*Homarus americanus*) Postlarvae on Settlement”. In: *Fisheries Oceanography* 35.2, pp. 158–167.